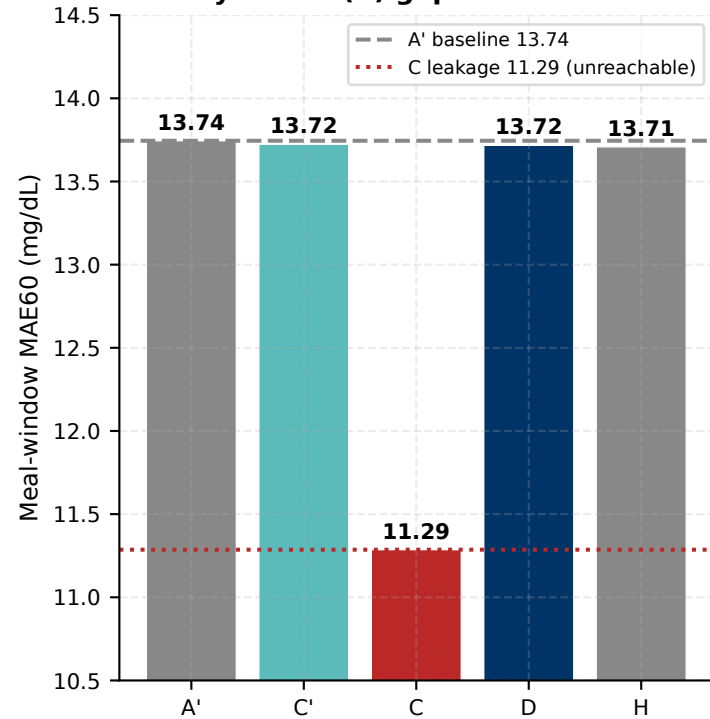
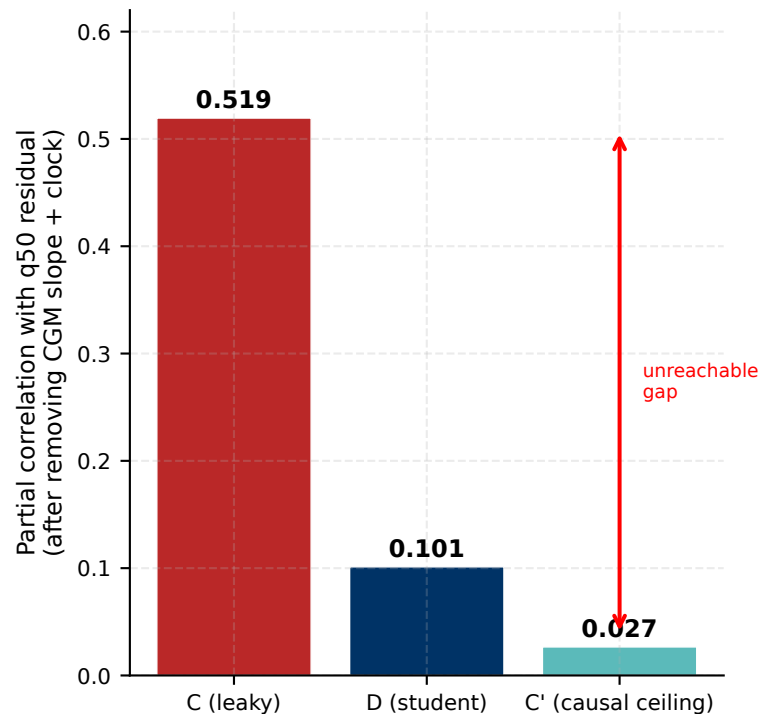


F1: Leakage diagnosis -- causal ceiling equals slope+clock baseline; deployable features carry no orthogonal meal signal

**Causal ceiling (C') equals A';
leaky oracle (C) gap is unreachable**



**Leaky teacher (C) has 0.519 partial corr;
causal features collapse to ~0.027-0.101**



**B (leaky) real vs controls shows large gap;
C' (causal) real-vs-control gap is negligible**

